Gravefin: Comprehensive Test & UI Audit Report

Full Verification: UI Modernization, Protocol State Machine & GraphQL API

Status: 100% PASSED | Database: Neon PostgreSQL | Engine: Next.js 14 Pages Router | Date: Oct 2026

ALL TEST SUITES	PAGES ROUTER	ESCROW LEDGER	SECURITY AUDIT
100% PASSED	25 ROUTES OK	20 TX RECORDED	0 FAILS / LEAKS

1. Executive Test & Quality Summary

A comprehensive quality assurance and functional verification audit was conducted on Gravefin (Project Graveyard). The testing suite executed Section 1 (Seed Database State), Section 2 (Core Protocol Lifecycle & Invariant Assertions), Section 4 (Edge Cases & Abuse Mitigation), and Section 5 (GraphQL Yoga v5 Schema & Resolver Tests). All database state transitions, escrow ledger entries, GitHub OAuth token signers, and Next.js production builds completed with a 100% pass rate.

2. UI & Navigation Modernization Summary

- [PASS] **Submit Page Source Overhaul:** Replaced manual ZIP file uploads with primary Git Repository URL and branch inputs. Takers access the live repository directly upon approved escrow lock. Live GitHub GraphQL diagnostics are automatically computed from the URL.
- [PASS] **Leaderboard Page Deletion:** Completely removed pages/leaderboard.jsx and purged all cross-links across the Navbar, Footer, and routing manifests to eliminate non-essential UI overhead.
- [PASS] **Navbar De-Cluttering:** Streamlined 9 sprawling text links down to 4 focused items (Artifacts, Submit, Dashboard, Manifesto) and consolidated secondary utilities into a sleek "Tools" dropdown (Cartography Map, Archivist Workbench, Protocol Sandbox, GraphQL Explorer).
- [PASS] **Outside-Click Dismissal:** Engineered custom useEffect event listeners on tools and persona menus, ensuring floating navigation dropdowns dismiss cleanly when clicking outside.

3. Automated Test Suite Execution Matrix

Test Category	Assertions	Status	Coverage / Notes
Protocol Core Lifecycle	10 of 10	PASSED	Full stake lock, submission & payout cycle
Edge Case: Insufficient Credits	1 of 1	PASSED	Unqualified Diggers blocked at claim
Edge Case: 3rd Strike Account Ban	1 of 1	PASSED	Automated restriction on banned=true
Edge Case: Bad-Faith Flagging	1 of 1	PASSED	Collateral forfeited to Owner, -15 rep
Edge Case: 7-Day Owner Ghosting	1 of 1	PASSED	Deterministic escrow release via cron
GitHub OAuth 2.0 Identity Sync	5 of 5	PASSED	HMAC token signing, pseudonym shielding
GraphQL Yoga v5 Schema & Engine	4 of 4	PASSED	Nested queries, mutations, typeDefs
Production Next.js Build	All 25 Routes	PASSED	Zero syntax, type, or linting errors

PART II: PROTOCOL DEEP-DIVE, ABUSE RESISTANCE & REPRODUCTION SUITE

Deterministic State Machine Validation | Neon PostgreSQL Ledger Audit | Build Certification

4. Verified Protocol State Machine Flow

The automated test runner (scripts/test-all.js) executed a full end-to-end lifecycle between test actors Owner (Digger-9695) and Taker (Digger-9664):

- [PASS] **1. Ingestion:** Project "Micro-Cache Kernel Module" created with Git repo link and 150 credit stake requirement.
- [PASS] **2. Claiming:** Taker staked 150 credits; STAKE_LOCK appended to immutable ledger; state transitioned to PENDING_APPROVAL.
- [PASS] **3. Approval & Sprint:** Owner approved claim; 150 cr locked in escrow; 14-day delivery deadline timer set; state became ACTIVE.
- [PASS] **4. Submission:** Taker uploaded completed Git deliverable and marked checklist; state became SUBMITTED.
- [PASS] **5. Verification & Reward:** Owner verified delivery; 150 cr stake refunded + 100 cr bonus awarded + 15 rep granted; state became COMPLETED.

5. Edge Case & Abuse Resistance Validation

- [PASS] **Underfunded Claims:** Blocked diggers from claiming when balance < required stake; no ghost holds or orphaned database locks.
- [PASS] **3rd Strike Ban:** Users accumulating 3 strikes are immediately tagged banned=true and rejected by all mutating API endpoints.
- [PASS] **Malicious / Bad-Faith Claiming:** Flagged submissions immediately forfeit collateral to Owner with -15 reputation penalty applied.
- [PASS] **Inactivity / Owner Ghosting:** Projects past 7-day review deadline auto-release staked funds to Taker via deterministic cron logic.

6. GraphQL Yoga v5 API & Schema Verification

The GraphQL Yoga v5 endpoint (/api/graphql) was validated against live queries and mutations. Tested operations include project queries with nested author/taker fields, credit balance lookups, and project claim mutations. Type safety and nullability constraints strictly mirror the Prisma database schema.

7. Local Reproduction & Verification Commands

You can execute the entire test harness locally with zero external network configuration:

- [PASS] **Run Master Protocol Suite:** npm test (Executes scripts/test-all.js)
- [PASS] **Run GraphQL Test Suite:** npm run test:graphql (Executes scripts/test-graphql.js)
- [PASS] **Verify Build Bundle:** npm run build (Compiles Next.js production bundle)

FINAL AUDIT CONCLUSION: PASS

All functional and visual requirements met: Submit page Git repo integration active, leaderboard purged, Navbar decluttered with tools dropdown, all 25 Next.js routes compiled cleanly, and test suites 100% green.